



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: B.E., IV Semester & ECE A

Course Code & Title: EC3401 & Networks and Security

Name of Faculty member (s): Ms.L.Krishna Kumari

Name of the Topic: RSA Algorithm

Name of the Innovative Practice: Mind Map

Date & Duration: 11.04.2023 & 15 Minutes

Description of Mind Map Activity:

Mind map is a tool that incorporates graphics, words, numbers, and color that represent and reflect a specific lesson or concept. It is a graphical way to represent ideas and concepts. Mind map is a visual thinking tool that helps structuring information, helping the students to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

Goals (Learning Outcomes):

The students will be able to illustrate various public key encryption techniques.

Use of appropriate methods:

Justification for choosing the topic using mind map activity: RSA Algorithm consists of various steps, equations and terms which is the very basic for encryption and decryption of messages. So to make the students to recall all the concepts, I have chosen this topic for conducting mind map activity since it is an easy way to make all the students to participate in this activity.

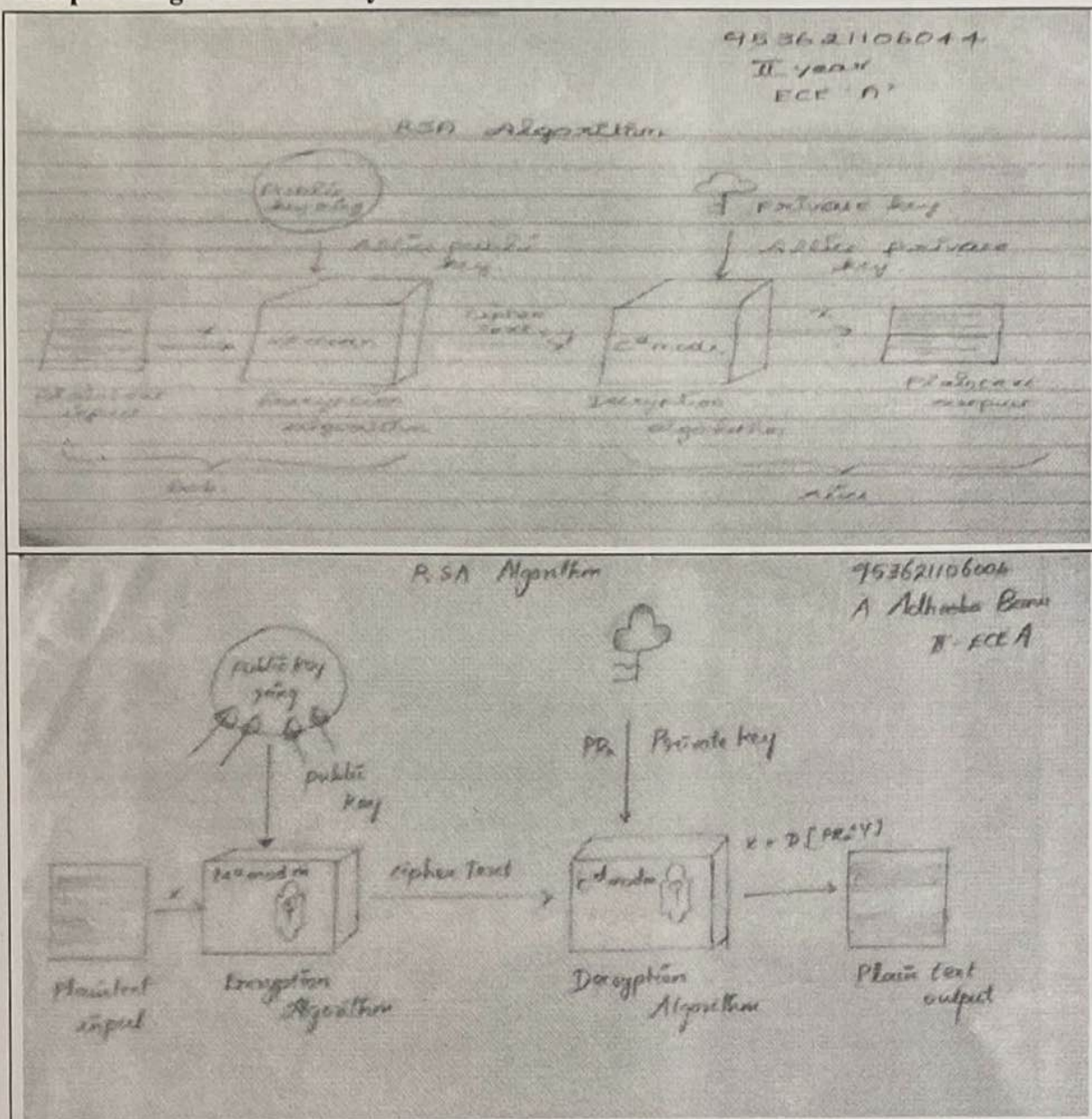
Effective presentation Details of the Implementation:

I called the students one by one to come to the stage and sketch the concept related to encryption and decryption of data using RSA Algorithm. So it was easy for them to recollect the points learnt in the classroom and they were able to view all the terms in a single activity. It was very useful for the students and they were also be able to reproduce the concepts in the exams. This mind map activity gives the graphical representation of words, concepts or pictures linked to and arranged around a given topic using a non-linear graphical layout which helps the students to build framework around a central concept.

Significance of Results:

Assessment of Effectiveness/Success of the Activity: By conducting this mind map activity, the students were able to recall the topics they had learnt in the classroom and they were also able to reproduce it in the exam. The assessment of effectiveness of the activity was felt when told most of the points.

Samples images of the activity:



Reflective Critique:

• Benefits of conducting this activity:

- ✓ It facilitates better understanding of relationships and connections between ideas and concepts of RSA Algorithm.
- ✓ This mind map activity helped the students to note down the most important terms or information using key words and then make them connection between the facts and ideas visually keeping all of the thoughts together.

• Challenges:

Since all the students eagerly participated in this activity, I felt no challenge in conducting this activity

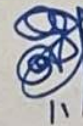
CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 8	PO 9	PO 10	PO 12	PSO 1	PSO 2	PSO 3
CO4	3	3	3	2	1	1	1	1	2	2	1	1

CO4: Determine the network security architecture and algorithms.

References:

1. Behrouz.A.Forouzan, Data Communication and Networking, Fifth Edition, TMH, 2017.
2. William Stallings, Cryptography and Network Security, Seventh Edition, Pearson Education, 2017.
3. Bhunia Swarup, Hardware Security –A Hands On Approach,Morgan Kaufmann, First edition, 2018.

 11/4/2023



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Innovative Practice Description

- **Unit / Topic:** V/Hardware Security/ Side channel attacks
- **Course Outcome:** CO5
- **Sub Topic:** 5.2
- **Activity Chosen:** Flipped Class room

- **Justification:**

The topic was chosen for flipped class because Side channel attacks and its counter measures. Unit 5 deals with Side channel attacks such as reverse engineering, probing and fault injection attacks and its counter measures. Since the students are exposed to different types of side channel attacks flipped class was given. This topic can be learnt in self-learning mode with simple manipulations and logic. The students have chance to browse the topic to know how the side channel can be carried out and how to overcome it. And also they find more information related to side channel attacks and circuits to mitigate this attacks. Thus this flipped class activity is chosen to make them to work out different techniques for side channel attacks.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group from side channel attack topic as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

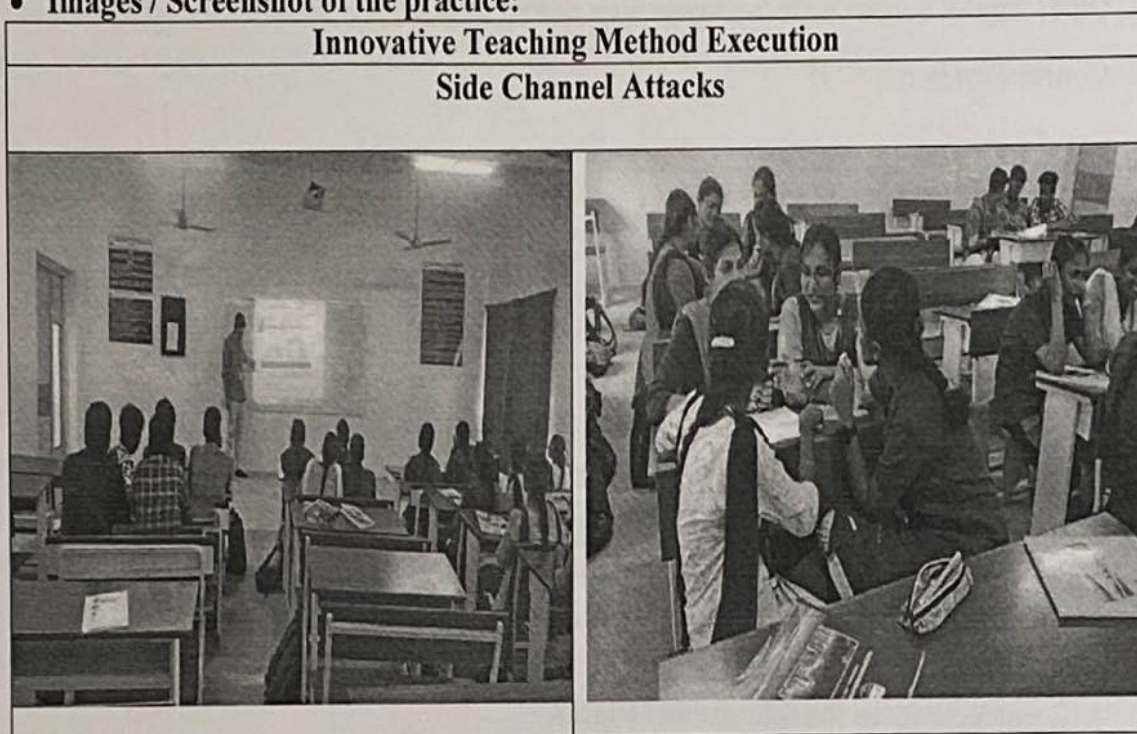
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 8	PO 9	PO 10	PO 12	PSO 1	PSO 2	PSO 3
CO5	3	3	3	2	1	1	1	1	2	2	1	1

- PO / PSO mapped

Innovative practice	PO2	PSO1
	3	2
Justification for correlation	To identify the various types of hardware security side channel attacks and provide suitable counter measures ; so course outcome is mapped with level 3	Students will be able to analyze various types of attacks related to hardware security and design a comprehensive and effective information and communication technology, hence course outcome is mapped with level 2.

- Images / Screenshot of the practice:



- Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ *Benefit of the practice:*

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of Buck, boost, buck boos analysis and design.

❖ *Challenges faced in implementation:*

Some of the students felt difficult to present because of lack of communication.

Few groups done their presentation beyond their time limit.

Few students in the group felt difficult to answer all the questions.

References:

1. Behrouz.A.Forouzan, Data Communication and Networking, Fifth Edition, TMH, 2017.
2. William Stallings, Cryptography and Network Security, Seventh Edition, Pearson Education, 2017.
3. Bhunia Swarup, Hardware Security –A Hands On Approach,Morgan Kaufmann, First edition, 2018.


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